Appendix –B (Inverse Kinematics of the robot, Leg 2 as stance leg)

Figure B.1 shows the inverse kinematics of the bipedal robot with leg 2 as the stance leg and leg 1 as the swinging leg. Using the cosine rule from trigonometry, it is obvious that

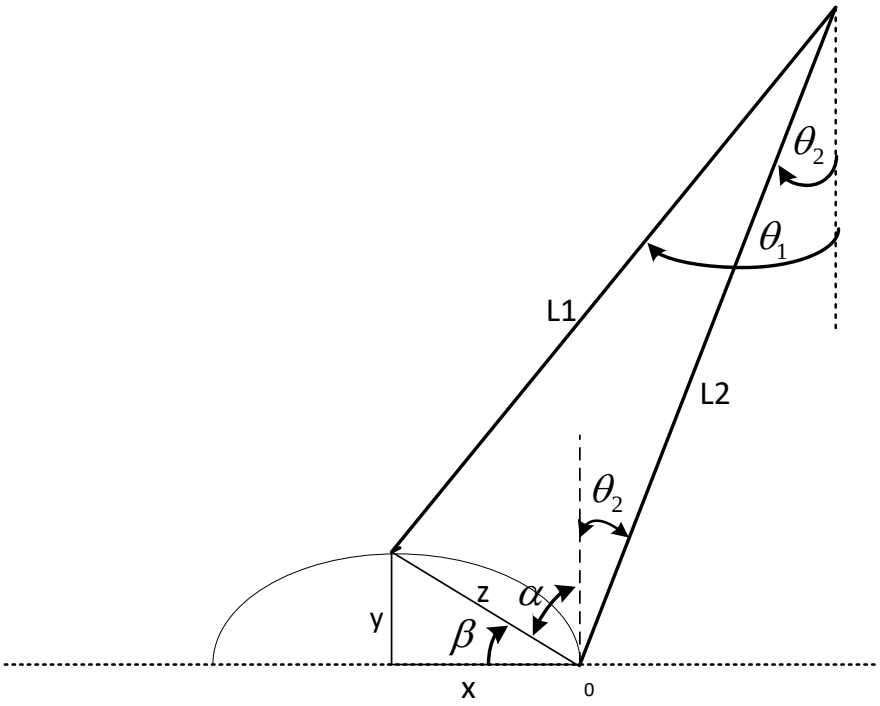


Figure B.1 Inverse Kinematics of the bipedal robot with leg 2 as the stance leg

$$L_1^2 = L_2^2 + z^2 - 2L_2\ z\ \cos\left(\alpha + \theta_2\right) \tag{B.1}$$

$$\theta_2 = \left(\frac{L_2^2 + z^2 - L_1^2}{2L_2\ z}\right) - \alpha \tag{B.2}$$

$$\alpha = \frac{\pi}{2} - \beta \tag{B.3}$$

$$\beta = \tan^{-1}\left(\frac{y}{x}\right) \tag{B.4}$$

$$\theta_1 = \cos^{-1}\left(\frac{L_2\cos\left(\theta_2\right) - y}{L_1}\right) \tag{B.5}$$

The horizontal and vertical components of the tip of leg 1 from the pivotal point o on the leg 2 are given as follows

$$y = -L_1 \cos(\theta_1) + L_2 \cos(\theta_2) \quad \text{(B.6)}$$

$$x = -L_2 \sin(\theta_2) + L_1 \sin(\theta_1) \quad \text{(B.7)}$$